12. A student was provided with five ionic solids containing familiar cations and anions. Each solid contains a different **cation**. The solids were labelled **A**, **B**, **C**, **D** and **E**.

The student attempted to dissolve all five solids in water. She then attempted to dissolve those that did not dissolve in water in a stoichiometric amount of acid.

The results are given below.

Solid	Addition of water	Addition of dilute sulfuric acid to solid	Addition of dilute nitric acid to solid
A	dissolves giving colourless solution **1**		
B	dissolves giving colourless solution **2**		
C	does not dissolve	pale blue solution **3** forms upon warming with the acid	
D	does not dissolve	effervescence and solution **4** forms	
E	does not dissolve	some effervescence but the solid does not dissolve	effervescence and solution **5** forms

Pairs of the solutions formed were mixed and the observations recorded.

	Solution **5**	Solution **4**	Solution **3**	Solution **2**
Solution **1** (formed by dissolving solid **A**)	no visible change	white precipitate	white precipitate	bright yellow precipitate
Solution **2** (formed by dissolving solid **B**)	no visible change	no visible change	brown solution with a white solid	
Solution **3** (formed by reacting solid **C** with acid)	thick white precipitate	no visible change		
Solution **4** (formed by reacting solid **D** with acid)	thick white precipitate			

Flame tests

Flame tests carried out on solutions **1**, **2**, **3**, **4** and **5** gave no colour with one solution, an apple-green flame with another and a lilac flame with a third. The student noted unfamiliar flame test colours for the other two solutions.

Examiner only

(a) Suggest which anion is present in substances **D** and **E**. Give a reason for your answer. [2]

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(b) Use **all** the information provided to suggest identities for compounds **A**-**E**.

Explain clearly how you identified compounds **A** and **C**. [6 QER]

A ....

B ....

C ....

D ....

E ....

Explanation

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END OF PAPER

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